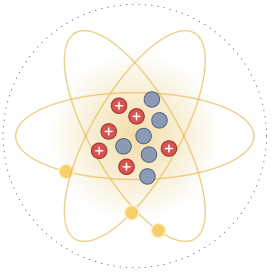
MT Classes, Motihari

Journey Inside the Atom



Class 9 Science, Chapter 8 (NCERT Exploration, new syllabus)

Creative notes with 24 diagrams, every NCERT question solved, and a quick quiz.

Start Everything is made of atoms

A house and a human body look nothing alike. But zoom in far enough and both turn out to be built from the same tiny units: **atoms**. Atoms are so small that we can never see them with our eyes.

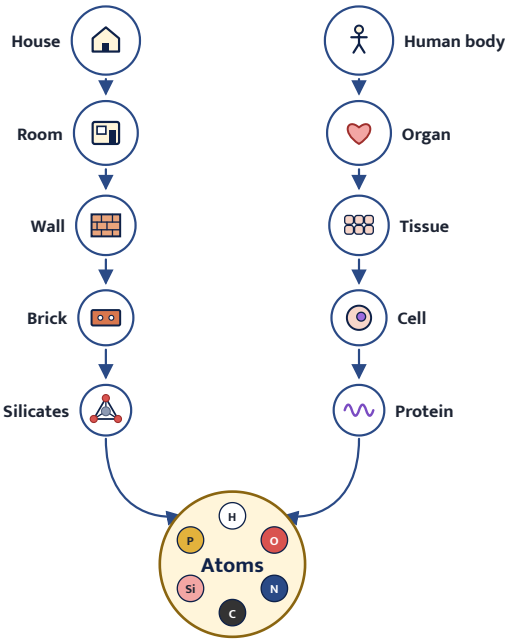


Diagram 1. Zoom into a house or a human body and you finally reach atoms (textbook chapter opener)

🤔 Big questions of this chapter

- Is an atom really the smallest piece of matter, or can it be broken further?
- Electrons are attracted by the protons in the nucleus. So why don't they fall into it?
- Why did scientists keep changing their picture (model) of the atom?

8.1 Where the idea of the atom began

More than 2,000 years ago, thinkers in India and in Greece asked the same question: **what is everything made of?**

Acharya Kanada Ancient India

Keep dividing matter (*dravya*) and you reach particles that cannot be divided further: **parmanu**. His ideas are recorded in the *Vaisheshika Sutras*.

Leucippus and Democritus Ancient Greece

Called the uncuttable particles **atomos** (Greek for "indivisible").

John Dalton 1808

First **scientific** atomic theory: all matter is made of tiny, indivisible atoms, the building blocks of matter.

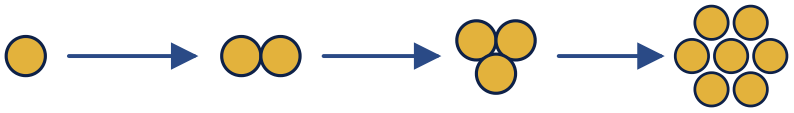


Diagram 2. Kanada's idea: parmanus combine into dyads, triads and so on, building all matter

- A parmanu is infinitely small and cannot be sensed. Their combinations make up the whole universe, including living bodies.
- **Gap in Kanada's idea:** it did not say in what **proportions** parmanus combine to form different substances.

💡 Key idea

The ideas of Kanada and Democritus came from **thinking and imagination**. Dalton's theory was the first one based on **experiments**, so it became the starting point of modern atomic structure.

After Dalton, scientists wanted to know three things: what are atoms made of, what would an atom look like, and what makes the atoms of one element different from another?

8.2 A journey through atomic models

A **model** is a picture that scientists imagine to explain what they observe. Every new experiment tested the model. When it failed, the model was improved. The early models were not fully correct, but they show how science moves forward one step at a time.

Clue: atoms can give out particles

Until the late 1800s, atoms were thought to be indivisible. Then scientists found that some elements give out invisible energy and particles on their own. This is called **radioactivity**, and what comes out is **radiation**. If particles can come out of an atom, the atom must have smaller parts inside it.

Discovery of the electron (J. J. Thomson, 1897)

- Thomson passed electricity through a gas at **very low pressure** in a glass tube with two electrodes, using a **high voltage**.
- Rays travelled from the **cathode (−)** to the **anode (+)**. These are **cathode rays**.
- Electric and magnetic fields showed the rays are streams of **negatively charged particles**, much lighter than atoms. These particles were later named **electrons**.
- The rays were the same whatever the cathode material or the gas in the tube. So **electrons are present in the atoms of every element**.
- Charge on an electron = -1.602×10^{-19} C. For convenience it is taken as **−1**.

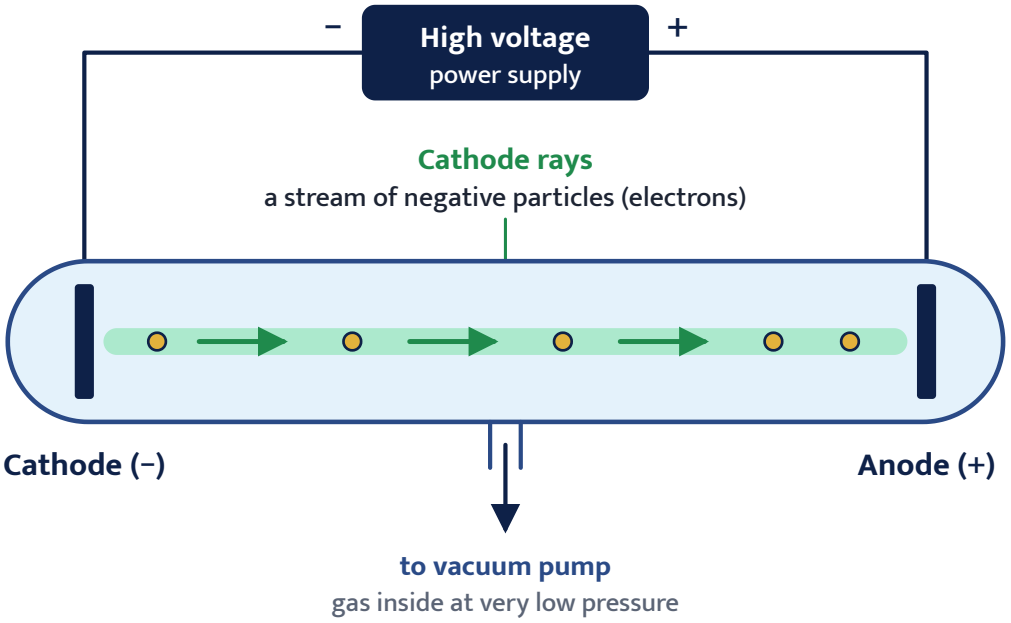


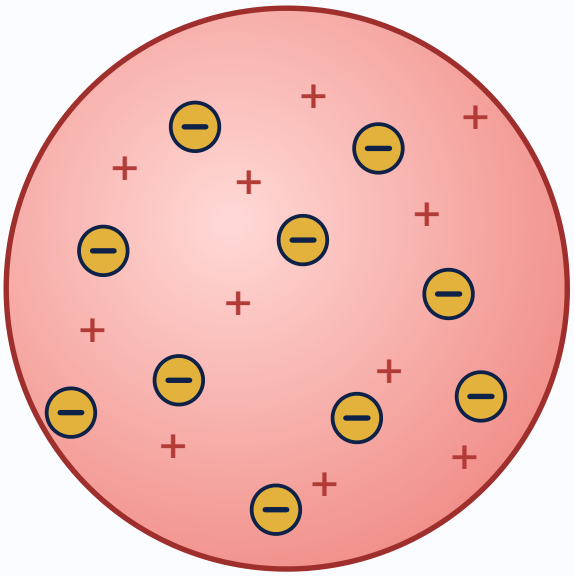
Diagram 3. Thomson's cathode ray tube (textbook Fig. 8.1)

Note: atoms have no colour. Colours in these diagrams only help you tell the parts apart.

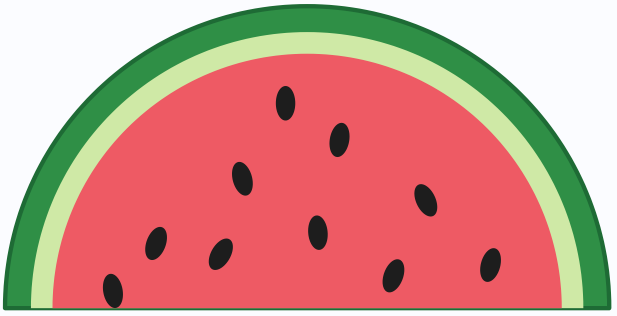
Thomson's model: the plum pudding (watermelon) model

The puzzle: atoms are neutral, but electrons are negative. So where is the positive charge?

- An atom is a **sphere of positive charge** with **electrons spread through it**, like plums in a pudding or seeds in a watermelon.
- Total positive charge = total negative charge, so the atom is **neutral**.



Thomson's model
A sphere of positive charge (+) with electrons (gold, −) spread through it



Like a watermelon
Red pulp = positive charge. Black seeds = electrons

Diagram 4. Thomson's model and the watermelon picture (textbook Fig. 8.2 and 8.3)

> 💡 **Why it matters**
>
> It was the first real attempt to show how the positive and negative charges in an atom balance each other. It was replaced when it failed to explain the gold foil experiment.

Testing Thomson's model: the gold foil experiment (1911)

Geiger and Marsden, working under **Ernest Rutherford**, fired a narrow beam of **alpha (α) particles** at an extremely thin gold foil.

- α-particles are tiny, **positively charged** particles given out by some radioactive elements. An α-particle is the **nucleus of a helium atom**: 2 protons + 2 neutrons.
- **Expected** (if Thomson was right): the positive charge is spread out, so α-particles should go straight through or bend only a little.

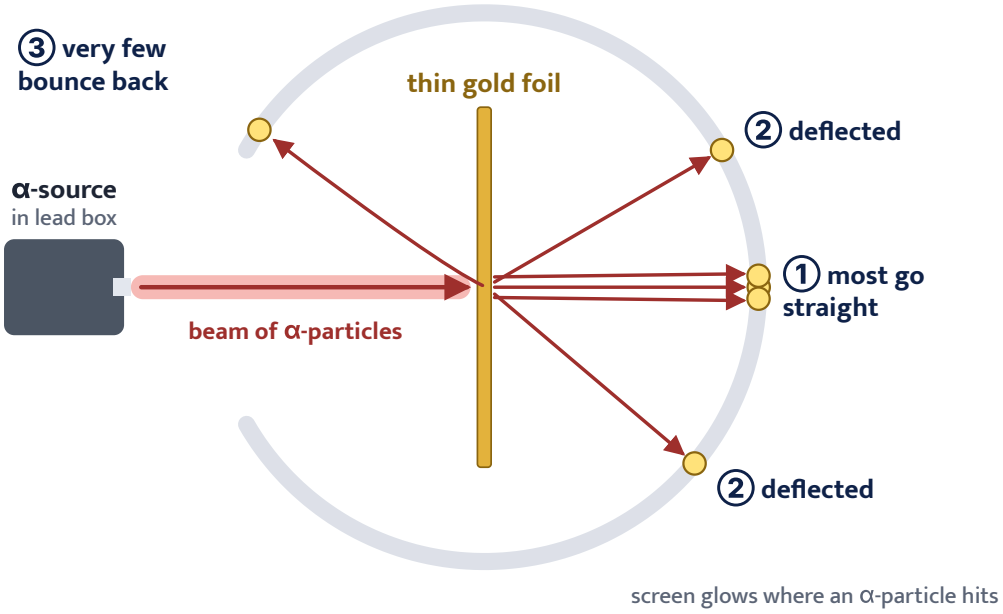
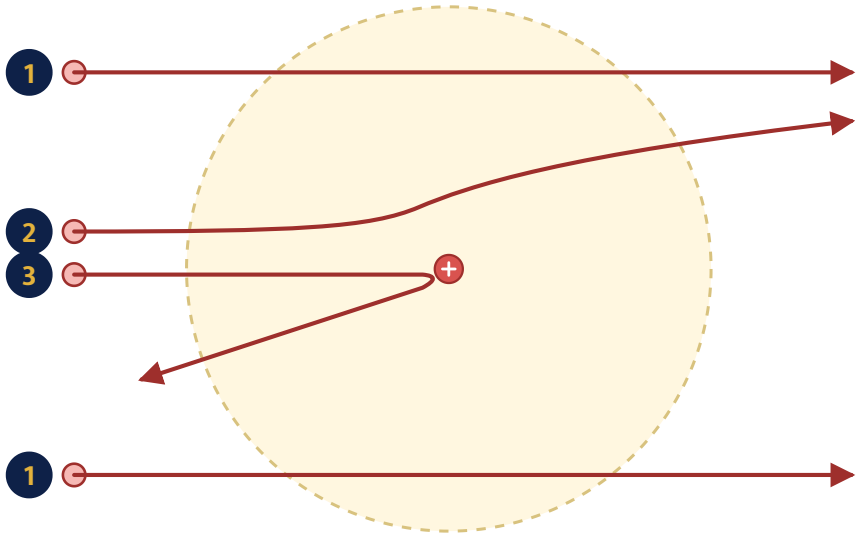


Diagram 5. Gold foil experiment: set-up and observations (textbook Fig. 8.4)

What they saw	What it means
① Most α-particles went **straight through**	Most of the atom is **empty space**
② Some were **deflected**, a few through large angles	The positive charge is not spread out. It sits in a **small region** that pushes α-particles away
③ Very few **bounced back**	That region is **tiny but very dense**: it holds almost all the mass



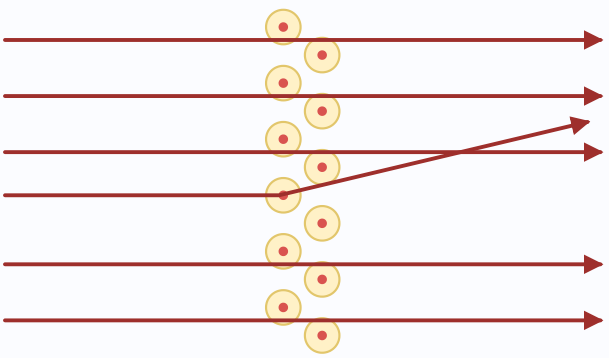
1. **Through empty space** (far from the nucleus): goes straight on. This happens to most α-particles.
2. **Close to the nucleus:** the positive nucleus pushes the positive α-particle away, so it is deflected.
3. **Aimed right at the nucleus:** a very strong push turns it back. This is rare because the nucleus is tiny.

Diagram 6. Zoom into one gold atom to see why each path happens

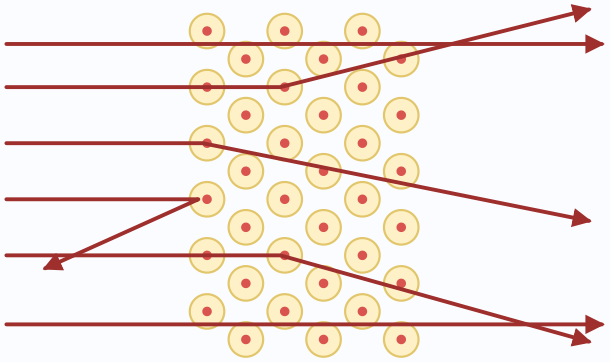
Turning of a particle away from its straight path is called **scattering**, so this is also called the **α-ray scattering experiment**.

🧪 Think as a scientist: what if the foil were thicker?

A thicker foil has more layers of gold atoms, so each α-particle has more chances of meeting a nucleus. **Fewer** would pass straight, and **more** would be deflected or bounce back.



Thin foil
Most α-particles pass straight through

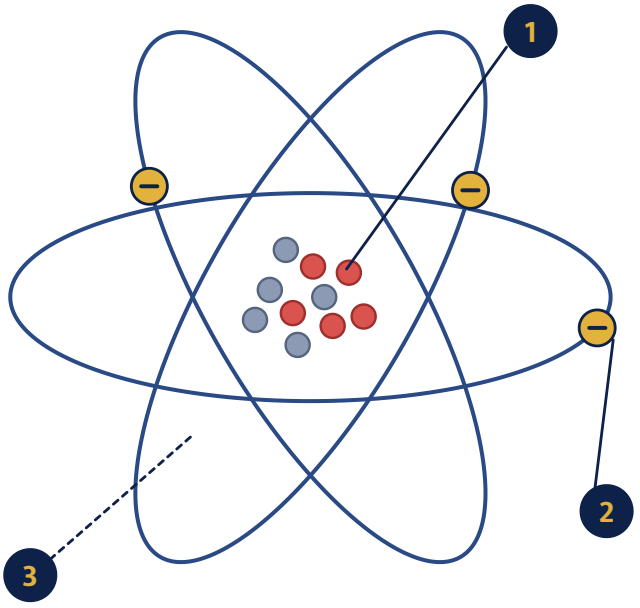


Thick foil
More are deflected, and more bounce back

Diagram 7. Prediction: thin foil versus thick foil

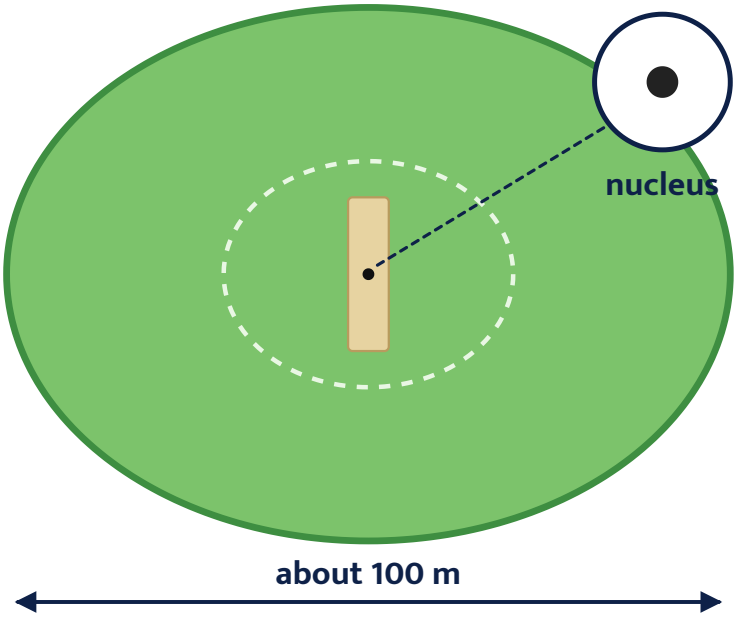
Rutherford's nuclear (planetary) model

- All the positive charge and almost all the mass is packed in a tiny centre called the **nucleus**.
- Most of the atom is **empty space**.
- Electrons **revolve around the nucleus**, like planets around the Sun. So it is called the **planetary model**.
- Size of an atom ≈ 10^{-10} m. Size of the nucleus ≈ 10^{-15} m. The nucleus is about **10^5 (one lakh) times smaller** than the atom.



1. **Nucleus:** tiny and dense, with all the positive charge and almost all the mass
2. **Electrons** revolve around the nucleus in orbits, like planets around the Sun
3. **Empty space:** most of the atom has nothing in it

Diagram 8. Rutherford's planetary model (textbook Fig. 8.5)



Atom	$\approx 10^{-10}$ m
Nucleus	$\approx 10^{-15}$ m

The nucleus is about **10^5 (one lakh) times** smaller than the atom. If an atom were a cricket ground, the nucleus would be a **pepper grain** (a few mm) at the centre.

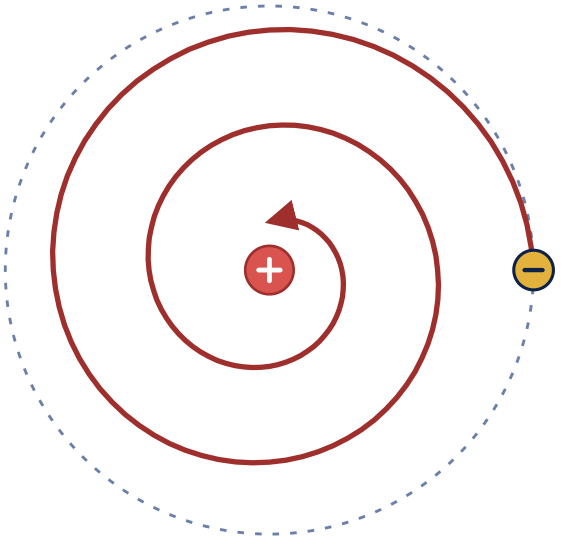
Diagram 9. Picture the size: a pepper grain in the middle of a cricket ground

✨ Try this calculation

A page of your book is about 0.1 mm = 10^{-4} m thick. One atom is about 10^{-10} m across. Atoms stacked across the page ≈ $10^{-4} \div 10^{-10} = 10^6$, that is about **ten lakh atoms**!

Where Rutherford's model went wrong

- Anything moving in a circle keeps changing direction, so it is **accelerating** (Chapter 4, Describing Motion Around Us).
- An accelerating charged particle should keep **losing energy**. The electron would spiral inwards and fall into the nucleus, and the atom would **collapse**.
- But atoms are **stable**; that is why matter around us stays intact. So the model could not explain the **stability of the atom**.



The atom would collapse!

A revolving electron is always accelerating, so it should keep losing energy, spiral inwards and fall into the nucleus. **But real atoms are stable.** Rutherford's model could not explain this.

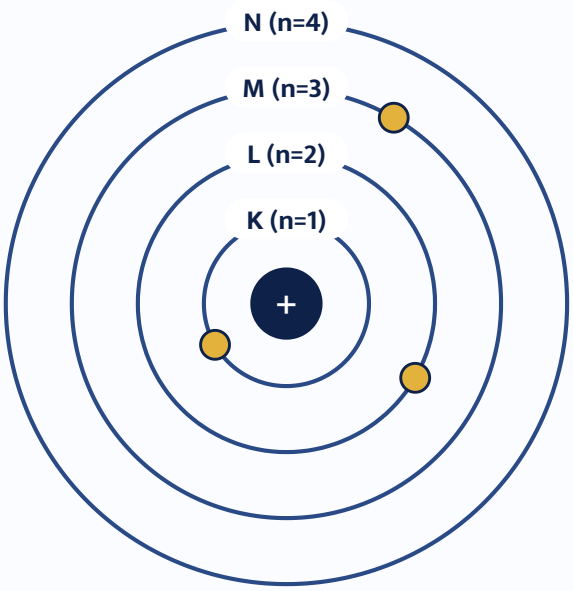
Diagram 10. Rutherford's problem: the electron should spiral into the nucleus (textbook Fig. 8.6)

Discovery of the proton

- Rutherford showed that the positive charge of the nucleus comes from particles he named **protons**.
- A proton carries a charge **equal and opposite** to an electron (+1) and is **much heavier** than an electron.
- In a neutral atom, **number of protons** = **number of electrons**. Helium: 2 and 2. Sodium: 11 and 11.

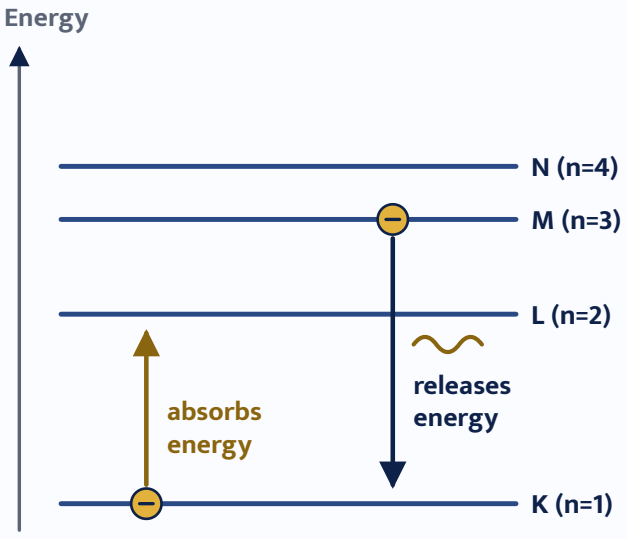
Bohr's model (Niels Bohr, 1913)

- Electrons move only in certain **fixed circular paths** called **stationary states**, orbits or **shells**.
- Each shell has a **definite energy**, so shells are also called **energy levels**. They are named **K, L, M, N…** or numbered **n = 1, 2, 3, 4…**
- An electron can be **in** a shell, never **between** shells. While moving in its shell, it **does not lose energy**.
- K (n = 1) is closest to the nucleus and has the **lowest energy**. Energy increases as you go outwards.
- To jump to a higher shell an electron **absorbs** energy; to drop to a lower shell it **releases** energy. The amount is exactly the energy difference between the two shells.
- Each shell can hold only a **limited number** of electrons.



Shells around the nucleus

K is closest; each shell has a fixed energy



Energy levels

Energy rises as you move away from the nucleus

Diagram 11. Bohr's shells and their energy levels (textbook Fig. 8.7)

💡 How Bohr explained stability

Bohr **assumed** (as a postulate) that in a stationary state the electron's energy stays constant even though it is moving. No energy loss means no spiralling in, so the atom is stable.

🔤 Why K, L, M, N and not A, B, C?

Physicist Charles Barkla named the first X-ray line he observed "K". He skipped A to J to leave room for any series found before K. None was ever found, and Bohr used the same letters for shells.

Bohr's model also has limitations. It was later replaced by the quantum mechanical model, which you will study in higher classes.

8.3 What gives an atom its mass?

Electrons are so light that their mass can be ignored. So the mass of an atom must be in its nucleus. But there was a puzzle:

🤔 The helium puzzle

Hydrogen has 1 proton and helium has 2 protons. So helium should be about **2 times** as heavy as hydrogen. Instead it is about **4 times** as heavy. Something else in the nucleus must be adding mass without adding charge.

Discovery of the neutron (James Chadwick, 1932)

- Chadwick, a student of Rutherford, found a particle with mass **nearly equal to a proton** but **no charge**. It was named the **neutron** (symbol **n**).
- Neutrons are found in the nucleus of **all atoms except ordinary hydrogen**.
- So the mass of an atom comes mainly from its **protons and neutrons**, packed tightly in the nucleus.

Particle	Symbol	Relative charge	Relative mass	Where	Found by
Electron	e^-	−1	about 1/1836 (negligible)	shells, around the nucleus	J. J. Thomson, 1897
Proton	p^+	+1	1	nucleus	Rutherford
Neutron	n^o	0	1	nucleus	James Chadwick, 1932

Why don't the protons in a nucleus fly apart?

All protons are positive, so each one pushes every other one away. Neutrons help in two ways: they sit between the protons and increase the distance between them (less push), and they add to the **nuclear**

force, a strong attraction that binds all the particles of the nucleus together.

Atom	Protons	Neutrons	Pattern
Carbon	6	6	light atoms: about equal
Oxygen	8	8	
Iron	26	30	heavy atoms: many more neutrons
Uranium	92	146	

✨ What if an atom had no empty space?

Everything would shrink enormously, because the nucleus is a lakh times smaller than the atom. If all the empty space were squeezed out of the atoms of every person on Earth, the whole of humanity would fit into roughly the size of a sugar cube!

⚛️ The atomic age

Having no charge, neutrons can enter nuclei easily. This led to man-made radioactive elements and to splitting uranium atoms, which gave us both nuclear power and nuclear weapons.

🇮🇳 India's contribution

The **Bhabha Atomic Research Centre (BARC)**, Mumbai, uses reactors like **Dhruva** for neutron-scattering experiments. This research helps develop superconductors, battery electrodes, medicines and industrial alloys in India. Find out more about Dhruva!

By 1869 scientists knew about 69 elements. Today we know **118** elements, some of them made artificially.

8.4 Symbols of elements

- **Dalton (1803)** was the first to use symbols: he drew **pictures** for elements.
- **Berzelius (1813)** suggested using **letters** taken from the Latin names of elements.
- Today, **IUPAC** (International Union of Pure and Applied Chemistry) approves the names and symbols of elements.

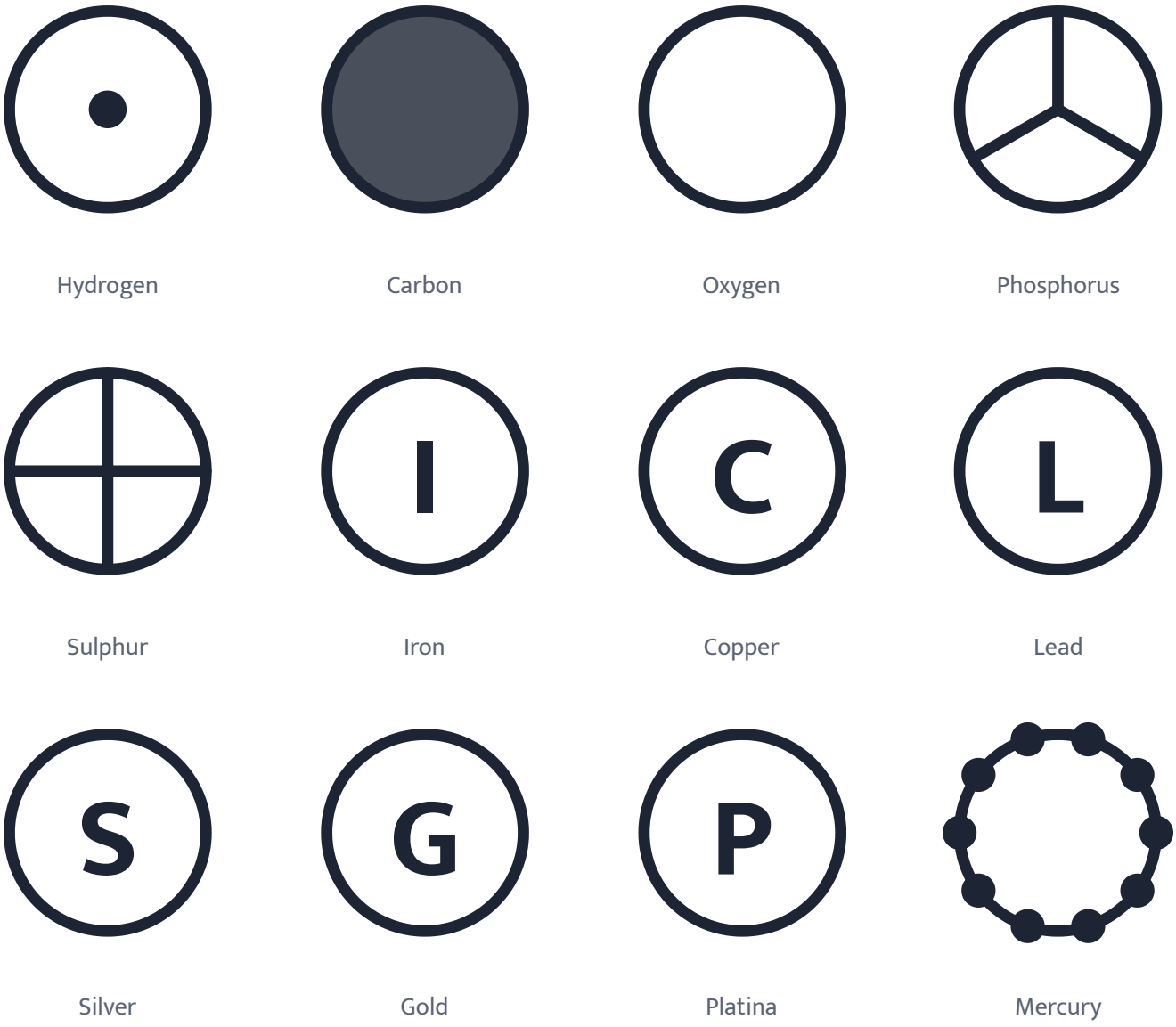


Diagram 12. Dalton's picture symbols for some elements (textbook Fig. 8.9)

Rules for writing symbols

1. Many symbols are the **first letter** or **first two letters** of the name: H, C, O, Al, Ca, Ne.
2. The **first letter is always capital** and the second letter is **small**: Al (not AL), Co (not CO).
3. Some use the first letter and a **later letter** of the name: Cl (chlorine), Zn (zinc), Mg (magnesium).
4. Some come from **Latin, Greek or German** names: Fe from *ferrum*, Hg from *hydrargyros* (Greek), W (tungsten) from *wolfram* (German).

> ⚠️ **Exam alert**
>
> **Co** is cobalt, an element. **CO** is carbon monoxide, a compound of carbon and oxygen. One capital letter changes everything!

Common elements and their symbols

Gold tiles: the symbol comes from the Latin name, shown in italics.

Al Aluminium	**Ar** Argon	**Ba** Barium	**B** Boron	**Br** Bromine	**Ca** Calcium	**C** Carbon
Cl Chlorine	**Co** Cobalt	**Cu** Copper *Cuprum*	**F** Fluorine	**Au** Gold *Aurum*	**H** Hydrogen	**I** Iodine
Fe Iron *Ferrum*	**Pb** Lead *Plumbum*	**Mg** Magnesium	**Ne** Neon	**N** Nitrogen	**O** Oxygen	**K** Potassium *Kalium*
Si Silicon	**Ag** Silver *Argentum*	**Na** Sodium *Natrium*	**S** Sulfur	**U** Uranium	**Zn** Zinc	

🧠 **Memory trick for the Latin symbols**

- **Fe**: *ferrum*, as in "ferrous" iron tablets
- **Cu**: *cuprum*, from Cyprus, an island famous for copper in ancient times
- **Ag**: *argentum*, the country Argentina is named after silver
- **Au**: *aurum*, think of a golden "aura"
- **Pb**: *plumbum*, the word "plumber" comes from it (old pipes were made of lead)
- **Na**: *natrium* and **K**: *kalium*

Symbols are used everywhere in the world, so scientists can understand each other whatever language they speak.

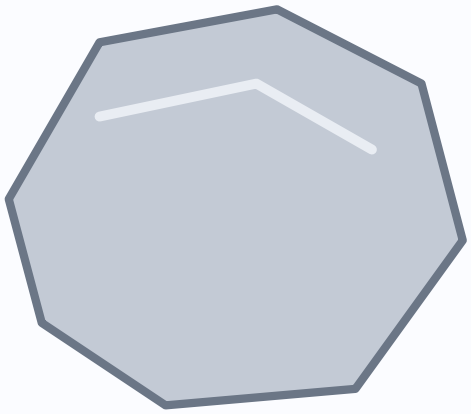
8.5 Atomic number and mass number

Atomic number (Z)

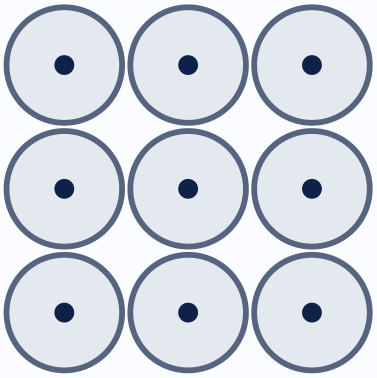
The **atomic number** is the **number of protons** in the nucleus of an atom. It is written as **Z**. It decides **which element** the atom is and how it behaves chemically.

- An atom is neutral, so **protons** = **electrons**. Hydrogen: Z = 1 (1 proton, 1 electron). Helium: Z = 2 (2 protons, 2 electrons).

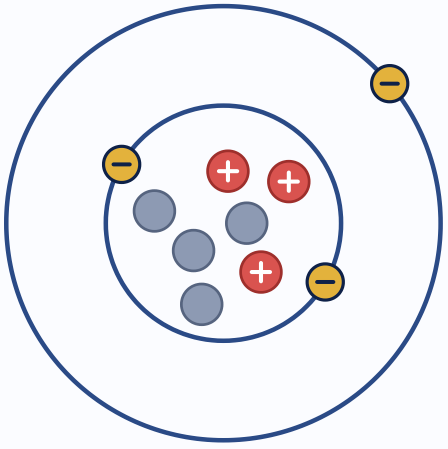
- Every element has its **own** atomic number. Change the number of protons and you get a different element.



A piece of lithium
made of...



lithium atoms.
Zoom into one...



One lithium atom
Nucleus: 3 protons + 4 neutrons. Electrons: 2 in K, 1 in L

proton (+) neutron (no charge) electron (−)

Diagram 13. Zooming into lithium: 3 protons and 4 neutrons in the nucleus, so Z = 3 (textbook Fig. 8.10)

Mass number (A)

The **mass number** is the **total number of protons and neutrons** in the nucleus. It is written as **A**. Protons and neutrons together are called **nucleons**. Electrons are so light that they are not counted.

Element	Protons (p^+)	Neutrons (n^o)	Mass number (A)
Hydrogen	1	0	1
Helium	2	2	4
Lithium	3	4	7

This solves the helium puzzle: 2 protons + 2 neutrons give helium a mass number of 4.

How to write an atom



Diagram 14. Standard notation: carbon has Z = 6 and A = 12

Z = number of protons = number of electrons

in a neutral atom

A = protons + neutrons

Neutrons = A − Z

✏️ Solved example

For $^{23}_{11}Na$: protons = 11, electrons = 11, neutrons = 23 − 11 = **12**.

For $^{35}_{17}Cl$: protons = 17, electrons = 17, neutrons = 35 − 17 = **18**.

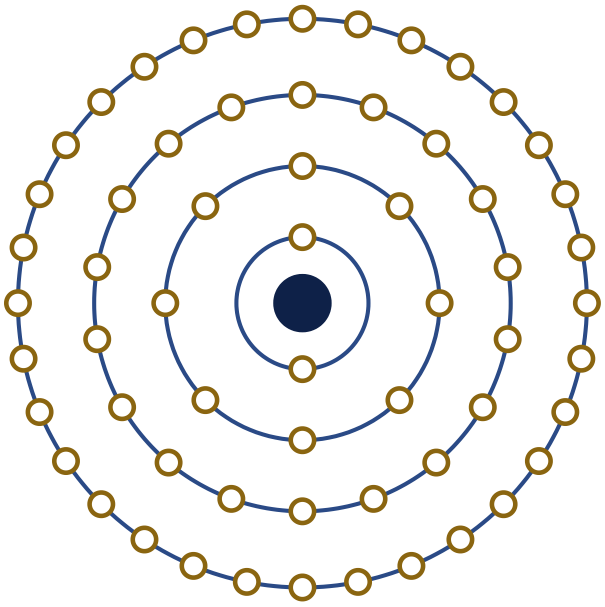
⚠️ Exam alert

Mass number is a **count** of particles, so it is always a whole number. Do not add electrons to it.

8.7 How electrons are arranged in shells

Bohr and Bury gave three simple rules for filling electrons into shells:

1. The **maximum** number of electrons a shell can hold is **$2n^2$**, where n is the shell number.
2. The **outermost** shell can hold **at most 8** electrons (the K shell alone holds at most 2).
3. Shells fill **step by step from the inside out**: K first, then L, then M, and so on. The L shell starts filling only after K is full.



Shell	$2n^2$	Max electrons
K (n = 1)	2×1^2	**2**
L (n = 2)	2×2^2	**8**
M (n = 3)	2×3^2	**18**
N (n = 4)	2×4^2	**32**

Diagram 15. Maximum number of electrons in each shell

The way electrons are distributed among the shells is called the **electronic configuration**. Hydrogen's single electron sits in K. Helium's two electrons both fit in K, which is now full. Each next element adds one more electron to the right shell.



Diagram 16. Electronic configuration of the first 18 elements. The gold ring is the outermost (valence) shell. (textbook Fig. 8.11)

Element	Z = p = e	n	K	L	M	Valency
Hydrogen (**H**)	1	–	1	–	–	**1**
Helium (**He**)	2	2	2	–	–	**0**
Lithium (**Li**)	3	4	2	1	–	**1**
Beryllium (**Be**)	4	5	2	2	–	**2**
Boron (**B**)	5	6	2	3	–	**3**
Carbon (**C**)	6	6	2	4	–	**4**
Nitrogen (**N**)	7	7	2	5	–	**3**
Oxygen (**O**)	8	8	2	6	–	**2**

Fluorine (**F**)	9	10	2	7	–	**1**
Neon (**Ne**)	10	10	2	8	–	**0**
Sodium (**Na**)	11	12	2	8	1	**1**
Magnesium (**Mg**)	12	12	2	8	2	**2**
Aluminium (**Al**)	13	14	2	8	3	**3**
Silicon (**Si**)	14	14	2	8	4	**4**
Phosphorus (**P**)	15	16	2	8	5	**3**
Sulfur (**S**)	16	16	2	8	6	**2**
Chlorine (**Cl**)	17	18	2	8	7	**1**
Argon (**Ar**)	18	22	2	8	8	**0**

The textbook asks you to add a valency column to this table. It is the last column above.

🧠 Remember the first 18 in order

Hi Hero, **Li**ttle **Be**e **B**uzzes **C**heerfully **N**ear **O**ld **F**lower **Ne**sts

H, He, Li, Be, B, C, N, O, F, Ne

Nana **Mag**ically **Al**ways **Si**ngs **P**retty **S**ongs **Cl**early **Ar**ound

Na, Mg, Al, Si, P, S, Cl, Ar

🤔 Think: why is potassium 2, 8, 8, 1?

Potassium has 19 electrons. The M shell can hold 18, so why not 2, 8, 9? Rule 2: the **outermost** shell cannot hold more than 8. So the 19th electron starts a new shell, N.

8.8 Combining capacity of an atom: valency

The **combining capacity** of an element is the number of hydrogen or chlorine atoms that **one atom** of it combines with. Hydrogen and chlorine are used as the measuring stick because each has a combining capacity of 1.

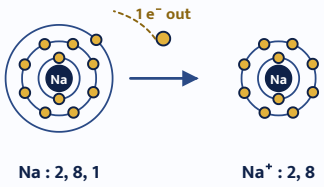
Compound	One atom of...	joins with	Combining capacity
H_2O (water)	oxygen	2 H	2

NH_3 (ammonia)	nitrogen	3 H	3
$MgCl_2$ (magnesium chloride)	magnesium	2 Cl	2
CH_4 (methane)	carbon	4 H	4

Where valency comes from

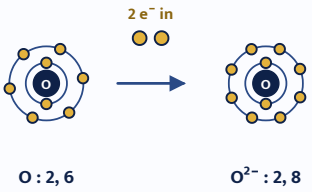
- The outermost shell that has electrons is the **valence shell**. Its electrons are the **valence electrons**.
- 8 electrons in the outermost shell is called an **octet**. Atoms with a complete octet (or 2 electrons, in helium's case) are **stable and largely unreactive**.
- Other atoms **lose, gain or share** electrons to complete their octet.
- **Valency** = the number of electrons an atom loses, gains or shares to complete its octet.

Lose



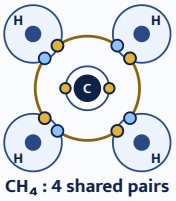
Sodium has 1 valence electron. It gives it away and is left with a full octet. Valency = 1

Gain



Oxygen has 6 valence electrons. It takes 2 more to complete its octet. Valency = 2

Share



Carbon has 4 valence electrons. It shares them, one with each of 4 hydrogen atoms (gold = carbon's, blue = hydrogen's). Valency = 4

Diagram 17. Three ways to reach a stable octet

Valence electrons	What the atom does	Valency	Examples
1, 2, 3	loses them	same as valence electrons	Na 1, Mg 2, Al 3
4	shares them	4	C, Si
5, 6, 7	gains electrons	8 − valence electrons	N 3, O 2, Cl 1
8 (He: 2)	nothing, already stable	0	He, Ne, Ar

Some compounds do not follow this simple rule (phosphorus, for example, can show valency 3 or 5). You will learn why in higher classes.

8.9 A deeper look: isotopes and isobars

Isotopes

Dalton believed all atoms of an element are identical and have the same mass. Scientists later found atoms of the same element with the **same number of protons** but a **different number of neutrons**.

> 💡 **Definition**
>
> **Isotopes** are atoms of the **same element** with the **same atomic number** but **different mass numbers**. Think of them as "twin atoms".

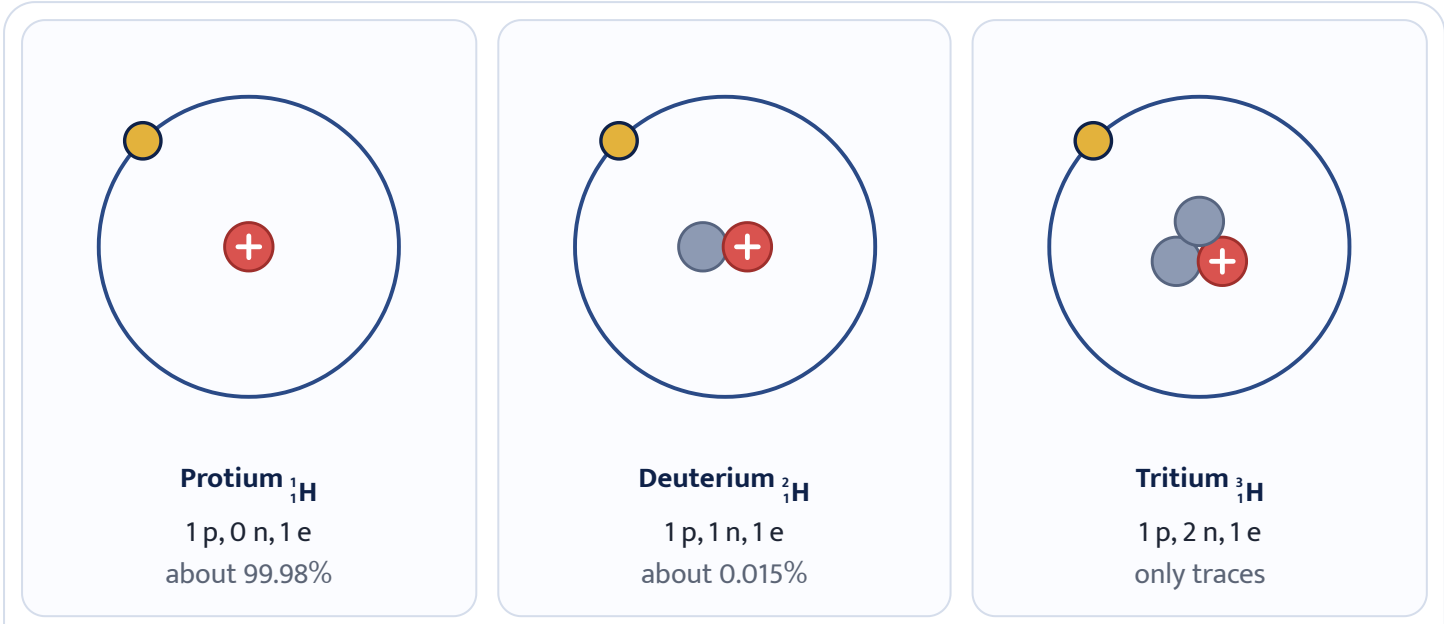
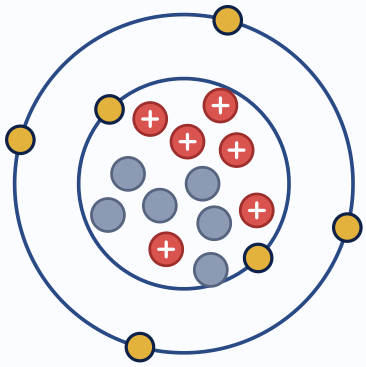


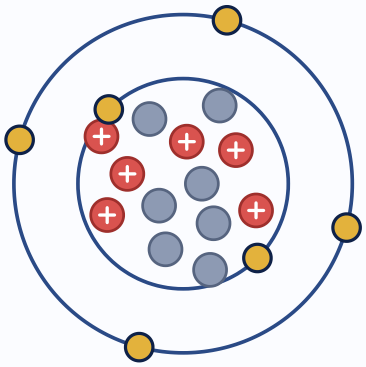
Diagram 18. The three isotopes of hydrogen. Each has 1 proton and 1 electron. (textbook Fig. 8.12)



Carbon-12 $^{12}_{6}C$

6 p, 6 n, 6 e

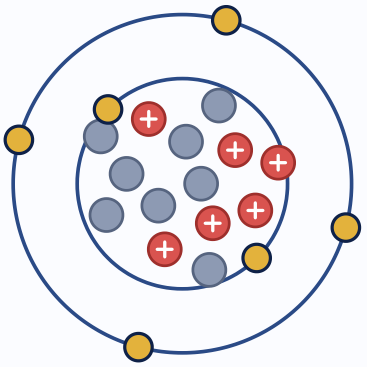
most abundant



Carbon-13 $^{13}_{6}C$

6 p, 7 n, 6 e

stable, rare



Carbon-14 $^{14}_{6}C$

6 p, 8 n, 6 e

radioactive: used in carbon dating

Diagram 19. The three isotopes of carbon. Each has 6 protons and 6 electrons. (textbook Fig. 8.13)

Property	Isotopes of an element	Why
Chemical properties	Same	Same number of electrons and same configuration; chemistry depends on valence electrons
Physical properties	Different (mass, melting and boiling points)	Different number of neutrons, so different mass

Isotopes at work

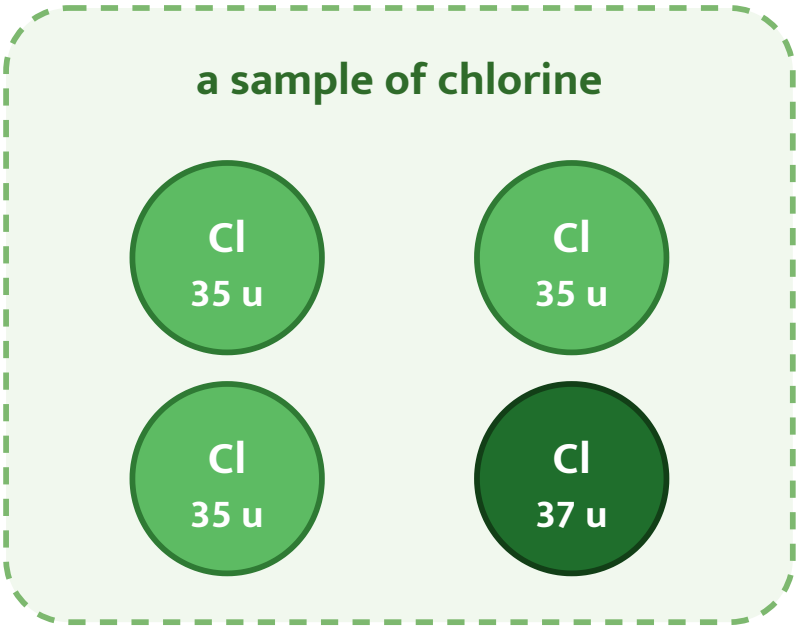
Isotope	Use
$^{235}_{92}U$ (uranium)	Fuel in nuclear reactors to produce electricity
$^{60}_{27}Co$ (cobalt)	Radiation treatment of cancer
$^{131}_{53}I$ (iodine)	Treatment of goitre and thyroid cancer
$^{14}_{6}C$ (carbon)	Finding the age of fossils and ancient objects

⚖️ Unit of atomic mass

Atoms are far too light to weigh in grams, just as you would not weigh a grain of wheat in kilograms. Atomic masses are measured in the **unified atomic mass unit (u)**. It was earlier written as amu.

Average atomic mass

Chlorine has two isotopes, ^{35}Cl (mass 35 u) and ^{37}Cl (mass 37 u), found in nature in the ratio **3 : 1**, that is 75% and 25%. So what is the mass of "a chlorine atom"?



Out of every 4 chlorine atoms, **3** are ^{35}Cl and **1** is ^{37}Cl (75% and 25%).

(35 + 35 + 35 + 37) ÷ 4 = 142 ÷ 4 = 35.5 u

Diagram 20. Weighted average: count each isotope as often as it occurs

Simple average = (35 + 37) ÷ 2 = 36 u

Wrong for nature: it treats both isotopes as equally common

Weighted average = 35 × 75/100 + 37 × 25/100 = 105/4 + 37/4 = 142/4 = 35.5 u

Correct: each mass is multiplied by its percentage abundance

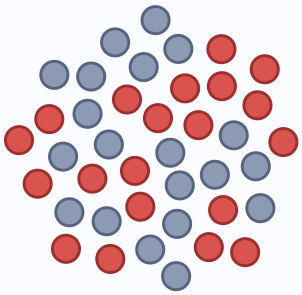
No single chlorine atom weighs 35.5 u. In 10 lakh chlorine atoms, about 7.5 lakh are ^{35}Cl and 2.5 lakh are ^{37}Cl, and their average comes to 35.5 u.

Isobars

Calcium (Z = 20), potassium (Z = 19) and argon (Z = 18) are different elements, yet each can have a mass number of 40: the total number of nucleons is the same.

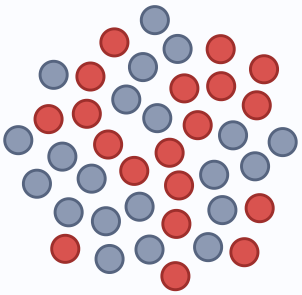
💡 Definition

Isobars are atoms of **different elements** with the **same mass number** but **different atomic numbers**.



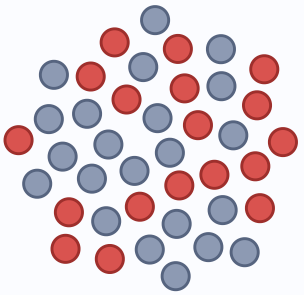
Calcium-40 $^{40}_{20}\mathrm{Ca}$

20 p + 20 n = 40



Potassium-40 $^{40}_{19}\mathrm{K}$

19 p + 21 n = 40



Argon-40 $^{40}_{18}\mathrm{Ar}$

18 p + 22 n = 40

Diagram 21. Isobars: three different nuclei, each with 40 nucleons

	Isotopes	Isobars
Element	same element	different elements
Atomic number (Z)	same	different
Mass number (A)	different	same
Neutrons	different	different
Chemical properties	same	different
Example	$^{35}_{17}\mathrm{Cl}$ and $^{37}_{17}\mathrm{Cl}$	$^{40}_{20}\mathrm{Ca}$ and $^{40}_{18}\mathrm{Ar}$

🧠 Never mix them up

IsotoPes have the same number of **P**rotons. IsobArs have the same **A** (mass number).

The journey is not over

Even Bohr's model is not fully correct. Electrons do not move in neat fixed orbits; they exist as **electron clouds** around the nucleus. We can predict where an electron is **most likely** to be, not exactly where it is. Today, electron microscopes such as the STM and TEM can even show individual atoms.

1



Dalton
1808
Atom is a tiny solid ball that cannot be divided

2



Thomson
after discovering the electron (1897)
Electrons stuck in a sphere of positive charge

3



Rutherford
1911
Tiny dense nucleus; electrons revolve around it

4



Bohr
1913
Electrons in fixed shells with fixed energy

5



Quantum model
1920s onwards
Electrons as a cloud: only likely regions

6



Still being discovered
today
Science keeps improving its models

Diagram 22. How our picture of the atom changed over time (textbook Fig. 8.16)



Revise in 10 minutes

Scientists of this chapter

AK **Acharya Kanada**
Ancient India. Idea of the *parmanu*, the smallest indivisible particle (Vaisheshika Sutras).

LD **Leucippus and Democritus**
Ancient Greece. Named the indivisible particle *atomos*.

JD **John Dalton**
1803 and 1808. Picture symbols for elements; first scientific atomic theory.

JB **Berzelius**
1813. Letter symbols taken from Latin names of elements.

JT **J. J. Thomson**
1897. Discovered the electron, the first subatomic particle; plum pudding model. Nobel Prize in Physics, 1906. As head of the Cavendish Laboratory, Cambridge, he guided Rutherford.

ER **Ernest Rutherford**
1911. Born in New Zealand. Gold foil experiment, the nucleus and the proton. Called the Father of Nuclear Physics. Nobel Prize in Chemistry, 1908. His face is on New Zealand's $100 note.

Niels Bohr

1913. Professor at Copenhagen University, Denmark. Fixed shells (energy levels) explained why atoms are stable. Nobel Prize, 1922.

Charles Barkla

Named X-ray lines starting from K, which is why shells are K, L, M, N.

James Chadwick

1932. Rutherford's student at the Cavendish Laboratory. Discovered the neutron. Nobel Prize in Physics, 1935.

Homi Jehangir Bhabha

Father of India's nuclear programme. Set up TIFR and BARC for peaceful uses of atomic energy: electricity, agriculture and medicine.

Quick revision

- Atoms are the building blocks of all matter, living and non-living.
- **Thomson:** electrons are embedded in a sphere of positive charge.
- **Rutherford:** the atom is mostly empty space with a tiny, dense, positive nucleus at the centre; electrons revolve around it. It could not explain stability.
- **Bohr:** electrons move in fixed shells (K, L, M, N...) without losing energy.
- Three subatomic particles: **electron** (−1), **proton** (+1), **neutron** (0, found by Chadwick).
- **Atomic number Z** = protons. **Mass number A** = protons + neutrons (nucleons).
- Shell capacity is $2n^2$; the outermost shell holds at most 8; fill K → L → M.
- An outermost shell with 8 electrons (2 for helium) is stable and largely unreactive.
- **Valency** = electrons lost, gained or shared to reach a stable octet.
- **Isotopes:** same Z, different A. **Isobars:** same A, different Z.
- **Average atomic mass** takes into account how common each isotope is in nature.

Must-know numbers

Charge on an electron	**-1.602×10^{-19} C** (taken as −1)
Size of an atom / nucleus	**10^{-10} m / 10^{-15} m** (nucleus about 10^5 times smaller)
Shell capacity ($2n^2$)	**K 2, L 8, M 18, N 32**
Neutrons	**A − Z**
Elements known	**69** by 1869, **118** today
Average atomic mass of chlorine	**35.5 u**

⚠️ Common mistakes to avoid

- Writing **CO** for cobalt. The symbol is **Co**.
- Adding electrons to the mass number.
- Confusing **mass number** (always a whole number) with **average atomic mass** (can be a decimal, like 35.5 u).
- Writing potassium as 2, 8, 9. The outermost shell cannot have more than 8, so it is 2, 8, 8, 1.
- Saying every nucleus has neutrons. Ordinary hydrogen has none.
- Saying the gold foil experiment found neutrons. It found the **nucleus**; neutrons came in 1932.
- Saying isotopes have different chemical properties. They are the **same**.

Quiz Test yourself

Tap an option. You will see the right answer and the reason straight away.

1. Who discovered the electron?

Ernest Rutherford	**J. J. Thomson**
James Chadwick	Niels Bohr

Thomson found electrons in 1897 while studying cathode rays.

2. Most α-particles passed straight through the gold foil. This shows that...

the nucleus is heavy	**most of the atom is empty space**
electrons revolve in orbits	neutrons exist

If most particles meet nothing, most of the atom must be empty.

3. The maximum number of electrons the M shell (n = 3) can hold is...

- 8
- **18**
- 32
- 2

$2n^2 = 2 \times 3^2 = 18$.

4. An atom has 17 protons and 18 neutrons. Its mass number is...

- 17
- 18
- **35**
- 1

A = protons + neutrons = 17 + 18 = 35.

5. The electronic configuration of sulfur (Z = 16) is...

- **2, 8, 6**
- 2, 6, 8
- 2, 8, 4, 2
- 8, 8

Fill K (2), then L (8), then the remaining 6 go to M.

6. The valency of nitrogen (2, 5) is...

- 5
- **3**
- 2
- 7

With 5 valence electrons it gains 3 more to complete the octet: 8 − 5 = 3.

7. Protium, deuterium and tritium are...

isobars	**isotopes**
different elements	noble gases

All three have 1 proton (same Z) but different numbers of neutrons.

8. Calcium-40 and argon-40 are...

isotopes	**isobars**
the same element	both noble gases

Same mass number (40), different atomic numbers (20 and 18).

9. Which model first explained why atoms are stable?

Dalton	Thomson
Rutherford	**Bohr**

Bohr's fixed shells, where electrons do not lose energy, explained stability.

10. The correct symbol for cobalt is...

CO	**Co**
Cb	co

First letter capital, second letter small. CO is carbon monoxide.

11. The average atomic mass of chlorine is...

36 u	**35.5 u**
35 u	37 u

Weighted average: 35 × 0.75 + 37 × 0.25 = 35.5 u.

12. Which particle has no charge?

Electron	Proton
Neutron	α-particle

The neutron is neutral; Chadwick discovered it in 1932.

NCERT Answers to the textbook questions

Questions are shortened. Tap a question to see its answer.

Pause and Ponder (in-text questions)

1 Thomson's atom made of clay (positive) and beads (electrons): what if (i) the clay's positive charge is less than the beads' total negative charge, (ii) the clay itself carries some negative charge?

(i) The negative charge would be more than the positive charge, so the model would have an overall **negative** charge. It would not be a neutral atom.
(ii) **No.** The clay's negative charge adds to the beads' negative charge, so total negative charge becomes more than total positive charge. A neutral atom needs the two to be exactly equal.

2 Is an orange or a lemon (seeds inside soft pulp) a good comparison for Thomson's model?

Where it matches: the seeds (electrons) sit inside a soft body (positive charge) that fills the whole fruit.
Where it falls short: the seeds are bunched in the centre and inside segments instead of being spread evenly; the fruit has a peel and segments that the model does not have; and nothing in a fruit carries charge, so it cannot show how charges balance.

3 Why did Thomson conclude that electrons are present in all atoms?

Cathode rays behaved the same whatever material the cathode was made of and whatever gas filled the tube. The same negative particles were coming out of every kind of atom, so electrons must be a part of all atoms.

4 What would happen if α-particles were replaced by negatively charged particles in the gold foil experiment?

The nucleus is positive, so negative particles would be **attracted** towards it instead of being pushed away. Most would still pass straight through the empty space. Those passing close to a nucleus would be pulled towards it and swing around it, so their paths would bend **towards** the nucleus.

5 How does the bouncing back of a few α-particles rule out the plum pudding model?

In Thomson's model the positive charge and mass are spread thinly over the whole atom. Such a spread-out charge can only nudge a fast, heavy α-particle a little; it can never stop it and throw it back. Bouncing back is possible only if the α-particle meets a **tiny region with a lot of positive charge and mass packed together**, that is, a nucleus. Thomson's model has no such region, so it must be wrong.

6 If you could ask Rutherford one question, what would it be?

Your own question is the answer here. Some ideas: "Why did you choose gold?" (Gold can be hammered into extremely thin sheets.) "How did you see where the α-particles went?" (Each one made a tiny flash of light on a zinc sulphide screen, and the flashes were counted.) "What did you think when you first saw particles bouncing back?"

7 Assertion: Rutherford concluded that most of the mass of an atom is in a small central region, the nucleus. Reason: In Thomson's model, electrons are embedded in a uniform sphere of positive charge.

Option (ii). Both statements are true, but the Reason describes Thomson's model, so it does not explain Rutherford's conclusion.

8 Name a new element after yourself and show that its symbol follows IUPAC rules.

Example: a student named Aarav could call it **Aaravium**, symbol **Av**. It follows the rules: the first letter is capital and the second is small, the letters come from the name, and no other element already uses Av. (A symbol such as Ar would not work: argon already has it.)

9 What problems could arise if every scientist used different symbols for the same element?

The same symbol could mean different elements to different people, so formulas and equations could not be shared or checked across countries. This would cause confusion and dangerous mistakes in labs, medicine and industry. A single IUPAC system avoids this.

10 An atom has atomic number 26 and 56 nucleons. Find its electrons, protons and neutrons.

Protons = 26, electrons = 26, neutrons = 56 − 26 = **30**. (This is iron.)

11 A nucleus has 20 protons and mass number 41. How many neutrons?

Neutrons = 41 − 20 = **21**.

12 An atom has 18 neutrons and atomic number 17. Find its mass number.

Mass number = 17 + 18 = **35**.

13 An atom with mass number 23 has 11 electrons. Find its neutrons.

Electrons = protons = 11, so neutrons = 23 − 11 = **12**.

14 Electrons in the outermost shell of carbon-12, fluorine-19 and silicon-28?

Carbon (Z = 6): 2, 4, so **4**. Fluorine (Z = 9): 2, 7, so **7**. Silicon (Z = 14): 2, 8, 4, so **4**.

15 Write the electronic configuration of elements with atomic numbers 12, 16 and 18.

Z = 12 (magnesium): **2, 8, 2**. Z = 16 (sulfur): **2, 8, 6**. Z = 18 (argon): **2, 8, 8**.

16 **Riddle: mass number 23, 11 protons, a soft metal that reacts vigorously with water. Who am I, and how many neutrons?**

Sodium (Na), with 23 − 11 = **12 neutrons**.
A riddle of your own: "I have 12 protons and 12 neutrons. I burn with a dazzling white light and I sit at the centre of chlorophyll. Who am I?" (Magnesium)

17 **Two atoms each have 11 protons; one has 12 neutrons, the other 13. Compare them.**

Atomic number: both **11** (same). Mass numbers: **23** and **24** (different). They are the **same element** (sodium), so they are **isotopes**.

18 **Bromine has two isotopes: bromine-79 (49.7%) and bromine-81 (50.3%). Find its average atomic mass.**

Average = 79 × 49.7/100 + 81 × 50.3/100 = 39.263 + 40.743 = **80.006 u ≈ 80 u**.

Other questions asked inside the chapter

★ **Combining capacity of nitrogen in NH_3 and of magnesium in $MgCl_2$?**

Nitrogen: **3** (joins 3 H). Magnesium: **2** (joins 2 Cl).

★ **How are helium's two electrons arranged?**

Both are in the **K shell**, which is then full (2).

★ **Lithium atom in the figure: protons, neutrons and atomic number?**

3 protons, 4 neutrons, so atomic number = **3** (and mass number = 7).

★ **How many electrons do protium, deuterium and tritium have?**

One each, because each has one proton.

★ **Will atoms that already have 8 outer electrons lose or gain electrons?**

No. Their octet is already complete, so they are stable and unreactive. Their valency is 0 (helium, neon, argon).

Revise, Reflect, Refine (exercises)

1 Gold foil experiment: which statements are correct? (i) It showed neutrons in the nucleus. (ii) It disproved the plum pudding model and led to a central nucleus. (iii) Large deflections showed that most mass and positive charge are in a tiny centre. (iv) The deflections showed that electrons move around the nucleus.

Correct: (ii) and (iii).

(i) Incorrect: neutrons have no charge and were found later, by Chadwick in 1932.

(ii) Correct: a spread-out positive charge cannot turn α-particles through large angles, so a central nucleus was needed.

(iii) Correct: only a tiny, heavy, concentrated positive centre can deflect fast α-particles so strongly.

(iv) Incorrect: the deflections were caused by the nucleus. Electrons are far too light to turn α-particles, so the experiment told nothing about how electrons move.

2 Bohr's model: correct or incorrect? (i) Electrons lose energy in fixed orbits and fall into the nucleus. (ii) Electrons can be anywhere with no fixed energy. (iii) Electrons revolve in orbits of fixed energy without losing energy. (iv) Electrons can be found between energy levels.

(i) **Incorrect:** in a fixed orbit an electron does not lose energy, so it never falls in.

(ii) **Incorrect:** electrons can be only in certain shells, each with a fixed energy.

(iii) **Correct:** this is Bohr's main idea.

(iv) **Incorrect:** an electron is never between levels; it jumps from one level to another by absorbing or releasing energy.

3 Nuclei: X (18 p, 19 n), Y (17 p, 18 n), Z (17 p, 20 n). Relation between (i) Y and Z, (ii) Z and X?

X: Z = 18, A = 37. Y: Z = 17, A = 35. Z: Z = 17, A = 37.

(i) Y and Z: same atomic number (17), different mass numbers (35 and 37), so they are **isotopes** (of chlorine).

(ii) Z and X: same mass number (37), different atomic numbers (17 and 18), so they are **isobars**.

4 What did Rutherford conclude about the positive part of the atom from the α-particles that bounced back or turned through large angles?

The positive part is at the **centre** of the atom, takes up a **very tiny** volume, and holds **all the positive charge and almost all the mass**, so it is extremely dense. He called it the nucleus.

5 Arrange the models in the order they were proposed and explain.

(iv) → (ii) → (iii) → (i)

Dalton (atoms indivisible) → Thomson (after the electron was found: electrons in a positive sphere) → Rutherford (gold foil experiment: dense central nucleus) → Bohr (fixed orbits to explain why atoms are stable). Each model fixed a problem in the one before it.

6 Why don't electrons fly away from the atom?

Electrons are negative and the nucleus is positive, so the nucleus **attracts** them. This pull towards the centre keeps them moving around the nucleus, just as the Sun's gravity keeps the planets in orbit. In Bohr's model they stay in their fixed shells without losing energy, so they neither fly away nor fall in.

7 Assertion: The discovery of subatomic particles helped in understanding atomic structure. Reason: An atom has equal numbers of electrons and protons.

Option (ii). Both are true, but equal numbers of electrons and protons is just one fact about atoms; it does not explain how the discoveries helped.

8 Magnesium atom with mass number 24 and atomic number 12: find protons, neutrons, electrons and draw its electron arrangement.

Protons = **12**, neutrons = 24 − 12 = **12**, electrons = **12**. Arrangement: **2, 8, 2** (K 2, L 8, M 2).

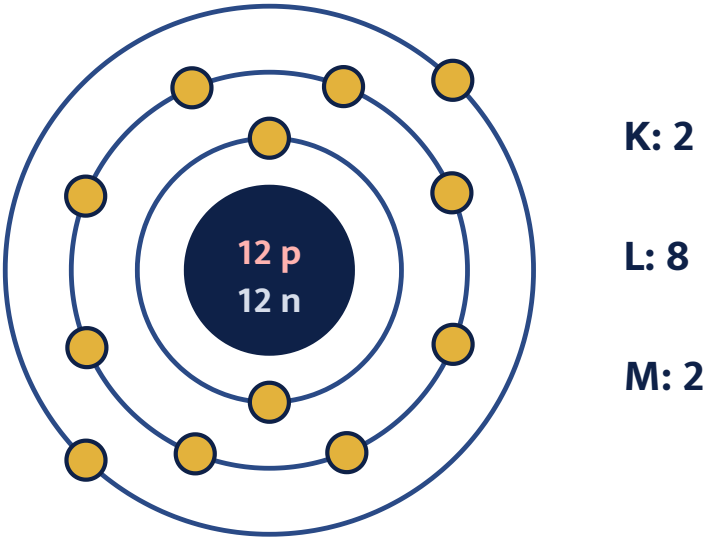


Diagram 23. Magnesium atom: 2, 8, 2

9 **For the four atoms shown in Fig. 8.17, find the name, symbol, total electrons, valence electrons, valency, protons and atomic number.**

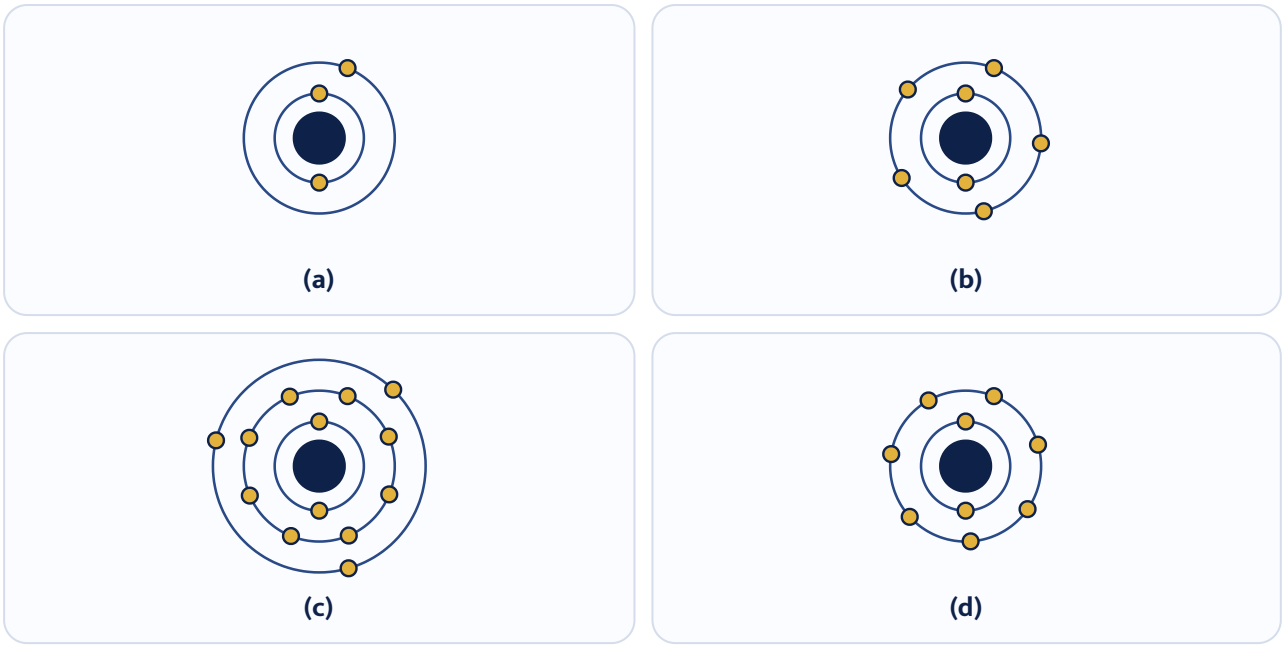


Diagram 24. The four atoms of Fig. 8.17

	Element	Symbol	Electrons	Valence e	Valency	Protons	Z
(a)	Lithium	Li	3 (2, 1)	1	1	3	3
(b)	Nitrogen	N	7 (2, 5)	5	3	7	7
(c)	Aluminium	Al	13 (2, 8, 3)	3	3	13	13
(d)	Fluorine	F	9 (2, 7)	7	1	9	9

10 **Both models have orbiting electrons. Why did Rutherford's fail to explain stability while Bohr's succeeded?**

In Rutherford's model nothing stops the revolving electron from losing energy. Because it is always accelerating, it should lose energy, spiral in and crash into the nucleus. Bohr **postulated** fixed stationary orbits in which the electron's energy stays constant; its energy changes only when it jumps between shells. With no energy loss, the atom stays stable.

11 **An atom with mass number 70 has 31 electrons. How many neutrons?**

Protons = 31, so neutrons = 70 − 31 = **39**.

12 **An atom has 79 protons and mass number 197. Find (i) neutrons, (ii) electrons.**

(i) Neutrons = 197 − 79 = **118**. (ii) Electrons = **79**. (This is gold.)

13 **Complete Table 8.5.**

Atomic number	Mass number	Neutrons	Protons	Electrons	Element
5	**11**	6	**5**	**5**	**Boron**
7	14	**7**	**7**	7	Nitrogen
12	24	**12**	12	**12**	**Magnesium**
15	**31**	16	**15**	**15**	**Phosphorus**
1	1	0	**1**	**1**	**Hydrogen**

14 **Element X has mass number 35 and 18 neutrons. Find (i) electrons and protons, (ii) atomic number, (iii) the element, (iv) configuration, (v) valence electrons, (vi) mass number if 2 neutrons are added, (vii) relation of X with the new atom.**

(i) Protons = 35 − 18 = **17**, electrons = **17**
(ii) Atomic number = **17**
(iii) **Chlorine (Cl)**
(iv) **2, 8, 7**
(v) **7** valence electrons
(vi) 35 + 2 = **37**
(vii) Same element with a different mass number, so they are **isotopes** (chlorine-35 and chlorine-37).

15 **An atom has 12 protons and 12 neutrons. If all its electrons were replaced by particles with the same charge but 500 times heavier, what happens to its (i) atomic number, (ii) atomic mass, (iii) mass number, (iv) overall charge?**

(i) **No change**: still 12 protons, so Z = 12.
(ii) **Increases.** An electron's mass is about 1/1836 u; 500 times that is about 0.27 u. Twelve such particles add about 3.3 u, so the mass rises from about 24 u to about 27.3 u.
(iii) **No change**: mass number counts only protons and neutrons, so it stays 24.
(iv) **No change**: 12 positive and 12 equal negative charges, so the atom stays neutral.

🔭 Explore more

- Fire α-particles yourself: PhET Rutherford Scattering simulation
- Build isotopes and see average mass change: PhET Isotopes and Atomic Mass simulation
- Project idea: draw a bar graph of electrons in each shell for any three elements.

✨ Something to wonder about

Can we ever understand **everything** that happens inside an atom? Every model so far has explained more than the one before it, and scientists are still exploring.

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Notes based on NCERT Exploration (Grade 9 Science), Chapter 8: Journey Inside the Atom. All diagrams are original illustrations made for these notes.